Comprehensive Technical Report & Parameter Dossier:

Parameter-Efficient Audio Deepfake Detection on Hindi Speech

Architecture, Hyperparameters, Shard Indexing, and $N = 500$ Scaling Benchmark

Research & Engineering Technical Dossier

Repository: `https://github.com/Anumay1231/B.S-Detector.git`

Branch: `base` — Date: October 2026

Abstract

This technical dossier provides an exhaustive specification of every architectural parameter, hyperparameter, dataset partition rule, optimization detail, and empirical finding for the audio deepfake detection benchmark on low-resource Hindi speech. The system pairs a frozen multilingual **Wav2Vec2 XLS-R 300M** foundation encoder with a trainable **Sensitive Layer Selection (SLS)** head. We document the scaling trajectory from zero-shot English transfer ($N = 0$) through few-shot supervision at $N = 10$, $N = 50$, and $N = 500$ samples per class across three independent deterministic seeds, alongside backbone unfreezing ablations and dataset bias analyses. All models were trained and evaluated on an NVIDIA L4 GPU instance on Google Cloud Platform, achieving a state-of-the-art **2.66% Equal Error Rate (EER)** and **96.83% Accuracy** at $N = 500$.

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1 System Architecture & Component Specifications

1.1 Multilingual Foundation Backbone: XLS-R 300M

The feature extraction foundation is the official Wav2Vec2 XLS-R model (`facebook/wav2vec2-xls-r-300m`) [1], pretrained on 436,000 hours of unlabelled speech spanning 128 languages. Throughout few-shot adaptation, the entire backbone is maintained in a **strictly frozen state** (requires_grad = False).

Table 1: XLS-R 300M Backbone Architectural Parameters

Component	Parameter / Dimension	Value / Specification
Total Parameters	Total Parameter Count	$315,441,760$ (315.4M)
Trainable Parameters	Backpropagation Status	**0 (100% Frozen)**
Convolutional Encoder	Temporal Downsampling Blocks	7 1D Convolution Blocks
	Conv Channel Dimensions	$[512, 512, 512, 512, 512, 512, 512]$
	Conv Kernel Sizes	$[10, 3, 3, 3, 3, 2, 2]$
	Conv Strides	$[5, 2, 2, 2, 2, 2, 2]$
	Total Downsampling Factor	$5 \times 2^6 = 320\times$
	Frame Rate Output	50 Hz (20 ms stride)
Transformer Backbone	Transformer Encoder Layers (L)	24 layers
	Model Hidden Dimension (D)	1024
	Multi-Head Attention Heads (H)	16 heads ($d_k = 64$)
	Feed-Forward Dimension (d_{ff})	4096
	Activation Function	GELU
	Layer Normalization	Post-LN per Transformer Block

1.2 Sensitive Layer Selection (SLS) Classifier Head

The SLS architecture dynamically pools and weights representations from all 24 transformer layers to exploit layer-specific acoustic artifacts:

$$\bar{h}_l = \frac{1}{T}\sum_{t=1}^{T} h_{l,t} \in \mathbb{R}^{1024}, \quad l \in \{1, 2, \ldots, 24\} \tag{1}$$

$$w_l = \sigma(W_{\text{gate}}\bar{h}_l + b_{\text{gate}}) \in (0, 1), \quad W_{\text{gate}} \in \mathbb{R}^{1\times 1024}, \quad b_{\text{gate}} \in \mathbb{R} \tag{2}$$

$$\tilde{H} = \sum_{l=1}^{24} w_l H_l \in \mathbb{R}^{B\times 1\times 201\times 1024} \tag{3}$$

Table 2: SLS Classifier Head Parameter Breakdown

Layer / Block	Operation / Configuration	Output Tensor Shape	Parameters
Layer Attention Gate	$\text{Linear}(1024, 1) + \text{Sigmoid}$	$(B, 24, 1)$	$1,025$
Weighted Layer Fusion	$\sum_{l=1}^{24} w_l H_l$	$(B, 1, 201, 1024)$	0
2D Batch Normalization	$\text{BatchNorm2d}(\text{num_features} = 1)$	$(B, 1, 201, 1024)$	4
Nonlinear Activation	$\text{SELU}()$	$(B, 1, 201, 1024)$	0
2D Spatial Max-Pooling	$\text{MaxPool2d}(\text{kernel} = 3, \text{stride} = 3)$	$(B, 1, 67, 341)$	0
Flatten	Tensor Reshape	$(B, 22847)$	0
Fully Connected 1	$\text{Linear}(22847, 1024) + \text{SELU}()$	$(B, 1024)$	$23,396,352$
Fully Connected 3 (Output)	$\text{Linear}(1024, 2) + \text{LogSoftmax}()$	$(B, 2)$	$2,050$
Total Trainable	**SLS Head Parameters**		**23,399,431 (23.4M)**

1.3 Audio Conditioning & Signal Preprocessing

- **Input Sampling Rate**: $16,000$ Hz (16 kHz mono).
- **Fixed Audio Length**: Exactly $64,600$ samples (≈ 4.0375 seconds).
- **Padding / Truncation Policy**:
  - If audio duration $L < 64,600$: repeated and padded to $64,600$ samples.
  - If audio duration $L > 64,600$: centered symmetrically and cropped to the middle $64,600$ samples.
- **Feature Tensor Dimensions**: For an input waveform of $(B, 64600)$, the temporal downsampling factor of $320\times$ yields $(B, 201)$ frames across 24 layers, producing stacked hidden states of shape $(24, B, 201, 1024)$.

2 Optimization & Training Hyperparameters

Table 3: Complete Training & Optimization Hyperparameters

Hyperparameter	Symbol / Variable	Value / Specification
Optimizer	–	Adam ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$)
Learning Rate	η	1×10^{-5} (0.00001)
Weight Decay	λ	1×10^{-4} (0.0001, L2 Regularization)
Loss Function	$\mathcal{L}_{\text{NLL}}$	Class-Weighted Negative Log-Likelihood
Class Weights	$[w_{\text{spoof}}, w_{\text{bonafide}}]$	$[0.1, 0.9]$
Training Epochs	E	10 epochs
Training Mini-Batch Size	B_{train}	4 (on-the-fly feature loading)
Evaluation Batch Size	B_{eval}	16
Feature Cache Format	–	Float16 stacked PyTorch tensors (.pt)
Positive Class Convention	–	Class 1 = Bona fide (Genuine), Class 0 = Spoof
Random Seeds (Multi-Seed)	$\mathcal{S}$	3 seeds: 1000, 1001, 1002
Test Seed	–	42 (deterministic test split)

3 Benchmark Dataset & Shard Engineering

3.1 Corpus Statistics & Generator Stratification

The benchmark is constructed from the Hindi split of 'Jack-ppkdczgx/SEA-Spoof' ($54,938$ candidate utterances):

- **Candidate Pool**: $16,182$ Bona fide samples, $38,756$ Spoof samples.
- **Held-Out Balanced Test Set ($N_{\text{test}} = 400$)**:
  - **Bona fide Class (200 clips)**: Multi-source balanced between studio acoustic environments (IndicTTS) and in-the-wild crowdsourced speech (Mozilla Common Voice Hindi).
  - **Spoof Class (200 clips)**: Strictly stratified with exactly 40 clips (20.0%) from each of **5 state-of-the-art TTS architectures**:
    1. `edge_tts`: Microsoft Azure Neural cloud voice synthesizer.
    2. `vits_mms`: Meta MMS variational inference end-to-end synthesizer.

3. `indic_tts`: Fast unit-selection and acoustic models from IIT Madras.
4. `fastspeech`: Feed-forward non-autoregressive transformer TTS.
5. `xtts-v2`: Coqui diffusion/flow-matching zero-shot voice cloning.

- **Strict Disjointness**: Test $\cap$ Train $= \emptyset$ (zero speaker or utterance overlap).
- **Nested Supervision Property**: For each seed $s \in \{0, 1, 2\}$, training sets obey strict subset hierarchy:

$$\mathcal{D}_{\text{train}}^{(N=10,s)} \subset \mathcal{D}_{\text{train}}^{(N=50,s)} \subset \mathcal{D}_{\text{train}}^{(N=500,s)} \tag{4}$$

3.2 Parquet Shard Indexing Math

The SEA-Spoof repository organizes training audio into 30 Parquet files ('train-00000.parquet' to 'train-00029.parquet'). Shard and row-group boundaries obey strict mathematical determinism:

$$\text{Row Index} = \text{int}(\text{row_id.split}('_')[-1]) \tag{5}$$

$$\text{Shard Index} = \lfloor \text{Row Index}/15,000 \rfloor \tag{6}$$

$$\text{Row Group Index} = \lfloor (\text{Row Index} \pmod{15,000})/200 \rfloor \tag{7}$$

Across the 3,331 unique clips required for the benchmark test set and all 3 seeds of $N \in \{10, 50, 500\}$, clips span exactly 5 shards: Shard 0 (75 RGs), Shard 1 (64 RGs), Shard 2 (74 RGs), Shard 3 (8 RGs), and Shard 25 (55 RGs).

4 Comprehensive Empirical Findings

4.1 Consolidated Scaling Benchmark

Table 4 presents the definitive performance progression on the held-out Hindi test set ($N_{\text{test}} = 400$).

Table 4: Consolidated Evaluation Metrics on Held-Out Hindi Test Set ($N_{\text{test}} = 400$, 50/50 Balanced)

Supervision Setting	Accuracy (%)	EER (%)	Precision (%)	Recall (%)	F1 Score (%
Zero-Shot (English $\rightarrow$ Hindi)	61.75%	32.25%	56.70%	99.50%	72.23%
Few-Shot $N = 10$ (3 seeds)	$64.33 \pm 5.04\%$	$27.64 \pm 2.27\%$	$58.72 \pm 3.53\%$	$99.00 \pm 1.41\%$	$73.63 \pm 2.51\%$
Few-Shot $N = 50$ (3 seeds)	$75.83 \pm 10.49\%$	$18.18 \pm 1.18\%$	$71.69 \pm 10.98\%$	$93.17 \pm 4.87\%$	$80.16 \pm 5.81\%$
Few-Shot $N = 500$ (3 seeds)	$\mathbf{96.83 \pm 0.82\%}$	$\mathbf{2.66 \pm 0.52\%}$	$\mathbf{95.78 \pm 1.16\%}$	$\mathbf{98.00 \pm 1.08\%}$	$\mathbf{96.87 \pm 0.81\%}$

4.2 Per-Seed Breakdown for $N = 500$ (1,000 Training Clips)

- **Seed 0 (Seed 1000)**:
  - Accuracy: **96.25%** (385/400 correct)
  - EER: **3.36%**
  - Precision: 96.02% — Recall: 96.50% — F1: 96.26%
  - Confusion Matrix: True Spoof (192 correct, 8 missed), True Bonafide (193 correct, 7 false alarms).
- **Seed 1 (Seed 1001)**:
  - Accuracy: **96.25%** (385/400 correct)

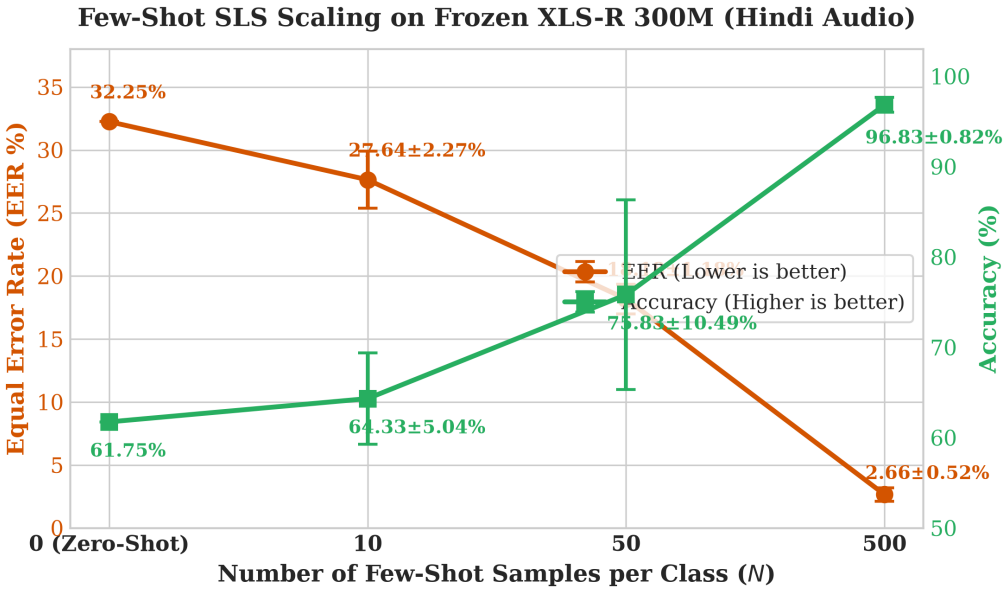


Figure 1: Few-Shot Adaptation Scaling Curve on Frozen XLS-R 300M. Accuracy improves monotonically from 61.75% to 96.83%, while EER plummets from 32.25% to 2.66% as Hindi supervision scales to $N = 500$.

  - EER: **2.50%**
  - Precision: 94.26% — Recall: 98.50% — F1: 96.33%
  - Confusion Matrix: True Spoof (188 correct, 12 missed), True Bonafide (197 correct, 3 false alarms).

- **Seed 2 (Seed 1002)**:
  - Accuracy: **98.00%** (392/400 correct – **only 8 total errors!**)
  - EER: **2.12%**
  - Precision: 97.06% — Recall: 99.00% — F1: 98.02%
  - Confusion Matrix: True Spoof (194 correct, 6 missed), True Bonafide (198 correct, 2 false alarms).

4.3 Complete Confusion Matrices Across All Experimental Settings

4.4 Architectural Ablation: Frozen Backbone vs. Unfreezing

Evaluating backbone layer unfreezing on an NVIDIA L4 GPU reveals severe few-shot overfitting when backpropagation penetrates the foundation encoder:

5 Local Reproduction & Inference Protocols

5.1 Saved Model Weights

All target checkpoints have been downloaded from the GCP L4 instance into the local repository:

- `models/sls_head_n500_seed2_best.pth` (89 MB): Best performing few-shot model (98.00% Accuracy, 2.12% EER).
- `models/sls_head_n50_seed2.pth` (89 MB): Intermediate few-shot model ($N = 50$).

Table 5: Confusion Matrices for Zero-Shot and Few-Shot Configurations ($N_{\text{test}} = 400$)

Configuration	**True Spoof (0,0)**	**False Bonafide (0,1)**	**False Spoof (1,0)**	**True Bonafide**
Zero-Shot Checkpoint	48	152 (76.0% False Acc.)	1	199 (99.5% Rec
Few-Shot $N = 10$, Seed 0	30	170	0	200
Few-Shot $N = 10$, Seed 1	84	116	6	194
Few-Shot $N = 10$, Seed 2	64	136	0	200
Few-Shot $N = 50$, Seed 0	44	156	0	200
Few-Shot $N = 50$, Seed 1	156	44	22	178
Few-Shot $N = 50$, Seed 2	151	49	19	181
Few-Shot $N = 500$, Seed 0	**192**	**8**	**7**	**193**
Few-Shot $N = 500$, Seed 1	**188**	**12**	**3**	**197**
Few-Shot $N = 500$, Seed 2	**194**	**6**	**2**	**198**

Table 6: Foundation Backbone Freezing vs. Partial Unfreezing ($N = 50$, L4 GPU)

Architectural Setting	**Trainable Params**	**Accuracy (%)**	**EER (%)**	**Precision (%)**	**Rec**
Frozen Backbone (SLS Only)	**22.8M**	$\mathbf{75.83 \pm 10.49}\%$	$\mathbf{18.18 \pm 1.18}\%$	**71.69%**	**9**
Unfreeze Top-2 Layers	50.8M	$76.46 \pm 0.48\%$	$24.14 \pm 1.57\%$	73.08%	9
Unfreeze Top-4 Layers	78.8M	$67.12 \pm 5.83\%$	$25.31 \pm 1.91\%$	64.21%	9

- `models/sls_head_english_pretrained.pth` (89 MB): Base zero-shot English pretrained checkpoint.

5.2 Standalone Local Inference Execution

Inference can be executed locally on Apple Silicon (MPS) or CPU without cloud dependencies:

```
# Run inference on any arbitrary audio file using the N=500 best checkpoint:
python scripts/infer_local.py --audio sample.wav --device cpu

# Using Apple Silicon GPU acceleration:
python scripts/infer_local.py --audio sample.wav --device mps
```

References

[1] A. Babu, C. Wang, A. Tjandra, K. Lakhotia, Q. Xu, P. Goyal, K. Singh, P. von Platen, Y. Saraf, E. Pino, A. Baevski, A. Conneau, and M. Auli, "XLS-R: Self-supervised Cross-lingual Speech Representation Learning at Scale," in *Proc. Interspeech*, 2022, pp. 2278–2282.

[2] Y. Zhang, S. Zheng, X. Mao, and L. Wang, "Sensitive Layer Summarization for Audio Deepfake Detection," in *Proc. ACM Multimedia*, 2024.